



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Jacob Conner
6/3/2026



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Objective: Predict first-stage landing success for a Falcon 9 rocket.

This is because being able to reuse boosters will reduce overall launch costs.

The approach included data collection, analysis, visualization, and machine learning.

Through this approach, we can use historical launch data to accurately predict landing outcomes.

Introduction

SpaceX is a private aerospace company lead by the belief in making space travel more affordable and working towards expanding human life into a multiplanetary species.

The Falcon 9 is the primary launch vehicle used by SpaceX. This is a two-stage partially reusable rocket that can be used for transport into orbit. One of the main benefits to the use of this rocket is the fact that the first stage can be recovered after the launch allowing it to be reused for the next launch. This greatly reduces the overall cost of a launch as well as the time required in preparation between launches.

From this research we ask the question, can we predict whether a Falcon 9 first-stage landing will be successful using historical launch data.

Section 1

Methodology

Methodology

Executive Summary

- Perform data collection
- Perform data wrangling
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models

Data Collection – SpaceX API

- <https://github.com/jacobconner069/Data-Science-Capstone-Project/blob/main/jupyter-labs-spacex-data-collection-api.ipynb>

Connect to SpaceX API ->
Request launch data ->
Retrieve launch records ->
Extract relevant records ->
Extract relevant fields ->
Make additional API calls ->
Merge retrieved information ->
Create DataFrame ->
Clean and transform data ->
Export dataset for analysis.

Data Collection - Scraping

- <https://github.com/jacobconnor069/Data-Science-Capstone-Project/blob/main/jupyter-labs-webscraping.ipynb>

Send HTTP request->

Retrieve Wikipedia HTML page ->

Parse HTML content ->

Locate launch records table ->

Extract table rows and columns ->

Clean and format data ->

Convert data to Pandas DataFrame ->

Handle missing values ->

Export clean dataset ->

Data Wrangling

- After the Data was collected, it was cleaned by removing missing and duplicate values, converting categorical variables to numerical, selecting only relevant features, and creating a target variable to be predicted.
- <https://github.com/jacobconner069/Data-Science-Capstone-Project/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>

Raw data sources ->
Import data into Pandas ->
Inspect dataset ->
Remove unnecessary features ->
Handle missing data ->
Convert data types ->
Create a target variable ->
Feature selection ->
Encode categorical variables ->
Clean and structured dataset.

EDA with Data Visualization

1. Flight Number vs Launch Success

Created to see if launch success improved over time.

2. Payload Mass vs Launch Success

Created to see if the mass of the payload influenced the success of the launch.

3. Launch Site vs Launch Success

Created to see if different launch sites had distinct launch success rates.

4. Orbit Type vs Launch Success

Created to see if rockets being sent into the same orbit had similar launch success rates.

EDA with Data Visualization Continued

5. Flight Number vs Payload Mass

Created to see if there was a change in payload mass over time

6. Pie Chart of Launch Outcomes by Site

Created to show the proportion of successful launches at each site.

7. Payload Range vs Landing Success

Created to show how changes in payload mass influence the success of the launch.

8. Interactive Geographic Map

Created to visualize the launch sites and the nearby area.

EDA with SQL

- Retrieved all unique launch sites
- Identified launch records from specific sites
- Counted the total successful landings
- Calculated successful landings by launch site
- Found launches with specific ranges of payload masses
- Determined missions with successful landings
- Identified outcomes of different orbits
- Identified launch sites with the most successful missions
- Examined relationships between payload mass and landing success

https://github.com/jacobconner069/Data-Science-Capstone-Project/blob/main/jupyter-labs-eda-sql-coursera_sqlite.ipynb

Build an Interactive Map with Folium

- Markers were added for each launch site with displayed names and coordinates to showcase the dispersion of launch sites throughout the country
- Circle markers were used to distinguish the launch sites from the rest of the map features
- Distance lines were drawn between launch sites and nearby points of interest to calculate distances from important features
- Markers were added to nearby points of interest such as coastlines, cities, railways, and highways to highlight features that may have an impact on the launch site capabilities
- Distance labels displayed the distances calculated by these distance lines from the launch sites to markers.

https://github.com/jacobconner069/Data-Science-Capstone-Project/blob/main/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

- The launch site dropdown menu allows users to select from the different launch sites to update the visualizations on the dashboard
- The launch success pie chart displayed the distribution of successful and failed launches over the user selected launch sites
- The payload range slider allowed users to select a specific range for the payload mass which was used to update other visualizations on the site
- The payload mass vs landing success scatter plot used the user selected values from the payload range slider to display similarities between the payload mass and the success of the launch.

https://github.com/jacobconnor069/Data-Science-Capstone-Project/blob/main/spacex_dash_app.py

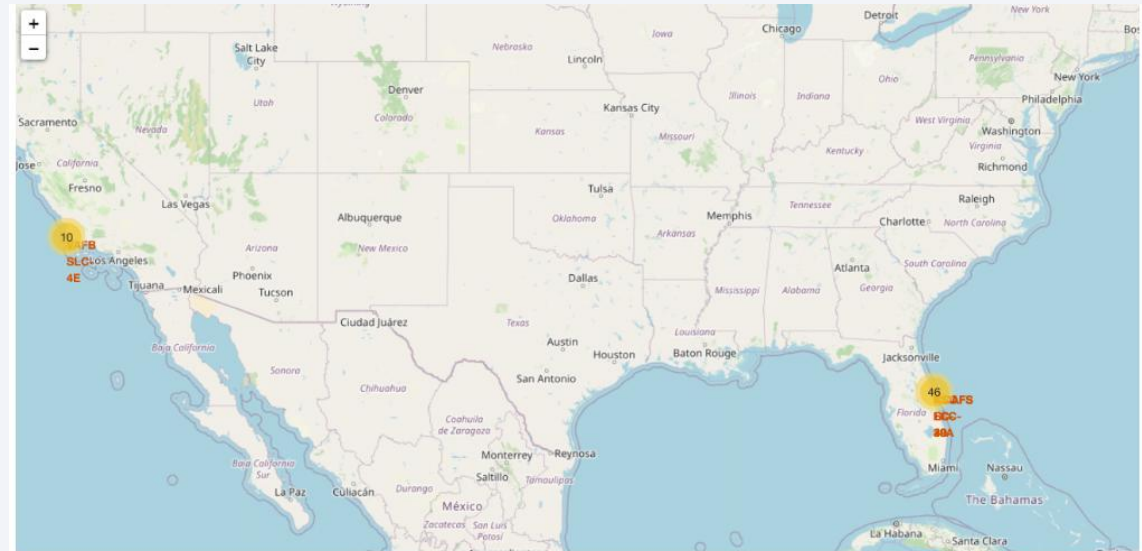
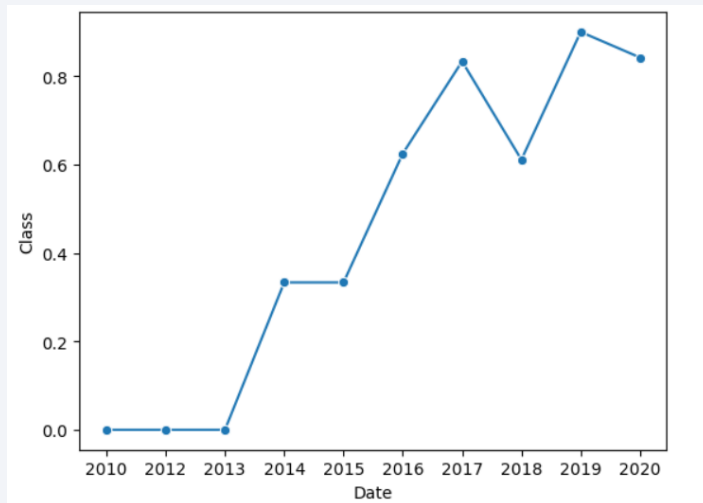
Predictive Analysis (Classification)

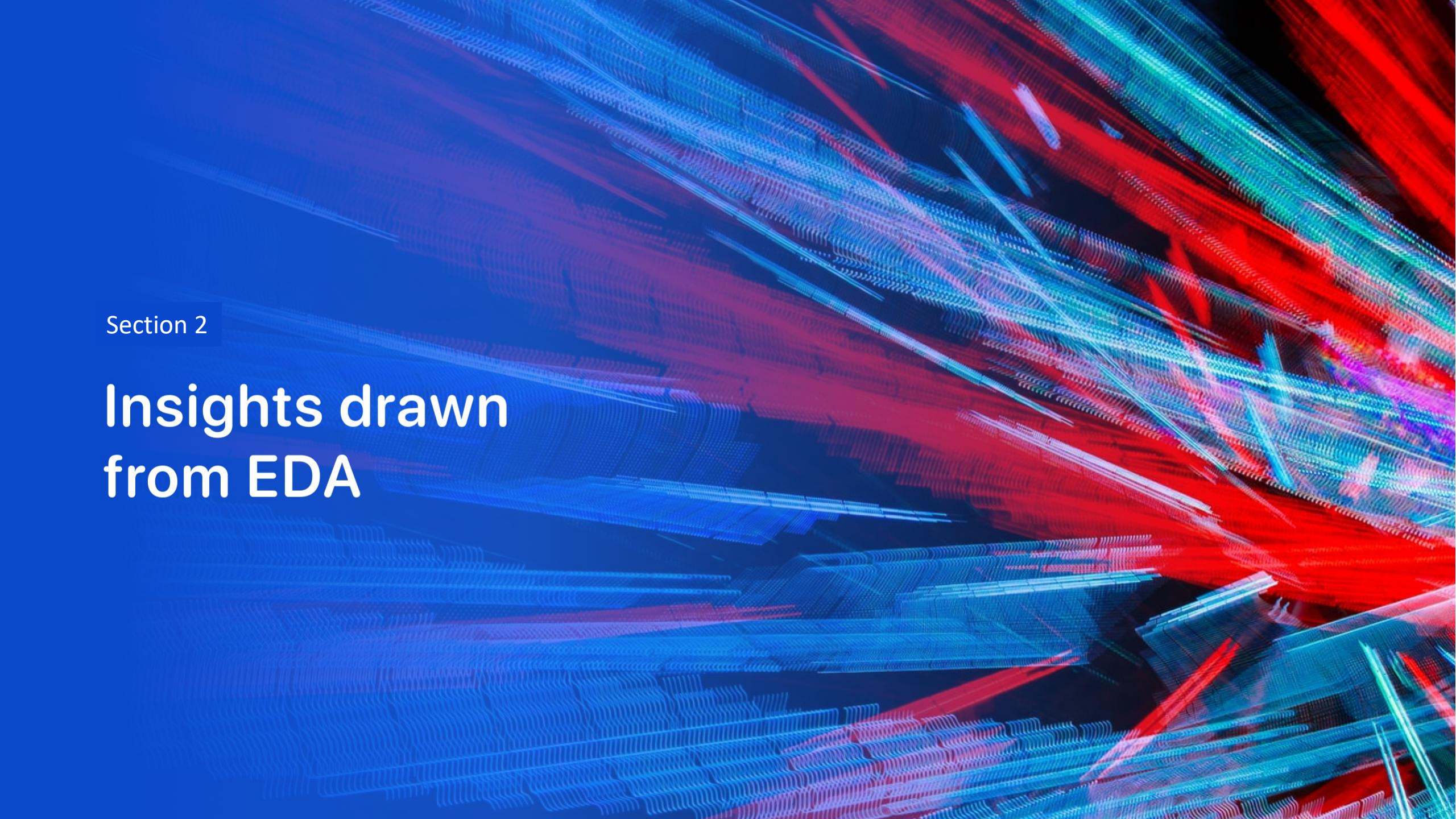
The models were built using the cleaned dataset which was standardized and split into training and testing sets. The models consisted of multiple classification algorithms, such as logistic regression, support vector machine (SVM), Decision tree, and K-Nearest neighbors (KNN). They were measured using accuracy on test data and cross validation was used for extra reliability. Different hyperparameters were tested for each algorithm and the parameters for the model with the highest accuracy was selected. Once the best model for each algorithm was selected the accuracy of each was compared so that the best classification model was used.

- Clean dataset ->
- Feature selection ->
- Data encoding ->
- Feature scaling ->
- Train-Test split ->
- Training classification models ->
- Hyperparameter tuning ->
- Model evaluation ->
- Compare results ->
- Select the best model ->
- Final prediction ->

Results

- Through EDA, we see that the success rate of the landings increased over time showing overall improvements in the design and capabilities of the rocket and the recovery process. Launch site, payload mass, and orbit type are the main factors influencing the success of the landing.
- Through predictive analysis, classification models were created with hyperparameter tuning and cross-validation showing that the K Nearest Neighbors algorithm was the most effective at predicting the outcome of the launch with an 83% accuracy rate.

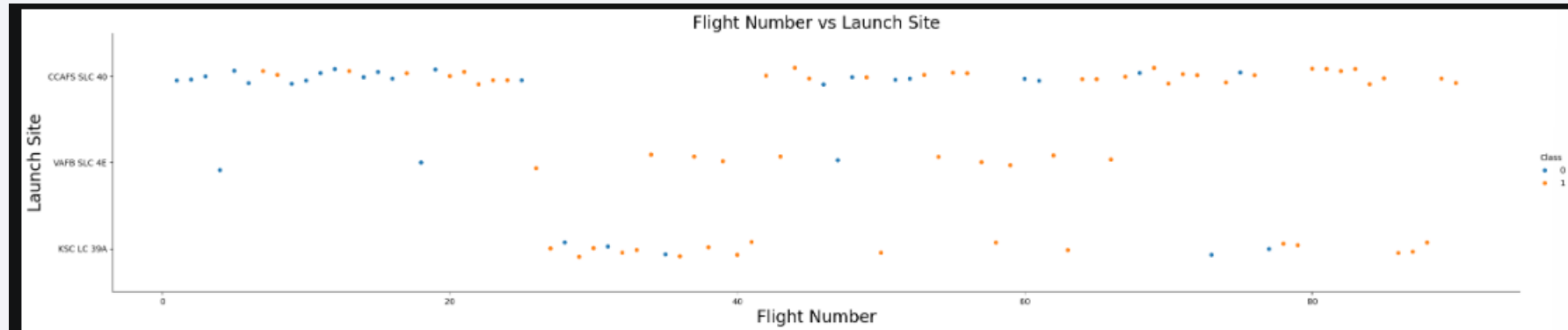


The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the upper right quadrant. The overall effect is dynamic and technological.

Section 2

Insights drawn from EDA

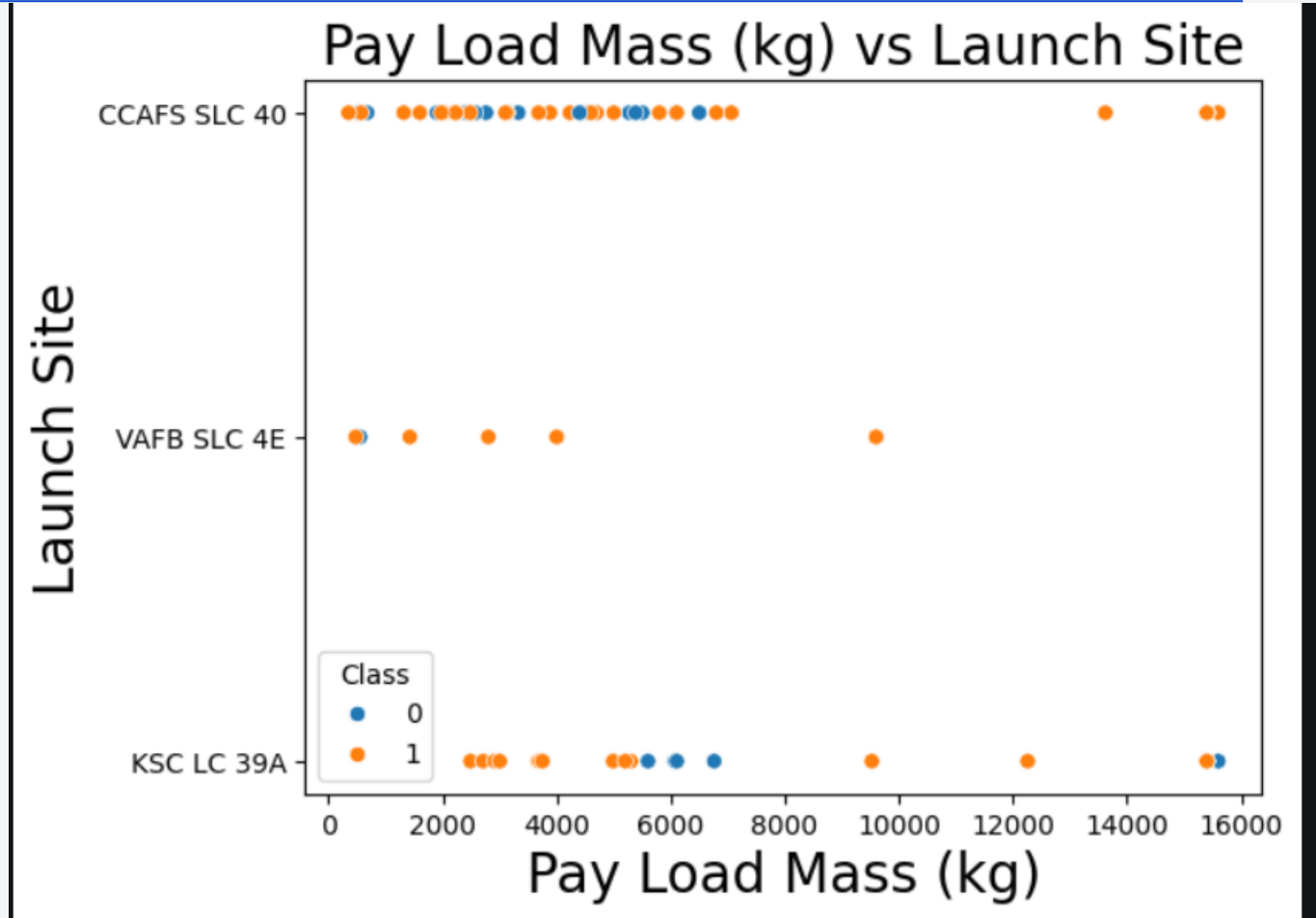
Flight Number vs. Launch Site



The scatter plot shows the progression of which launch sites were used over time. The flight number increases with each launch attempt and thus as time passes the flight number grows. This shows which launch sites were common in different time periods.

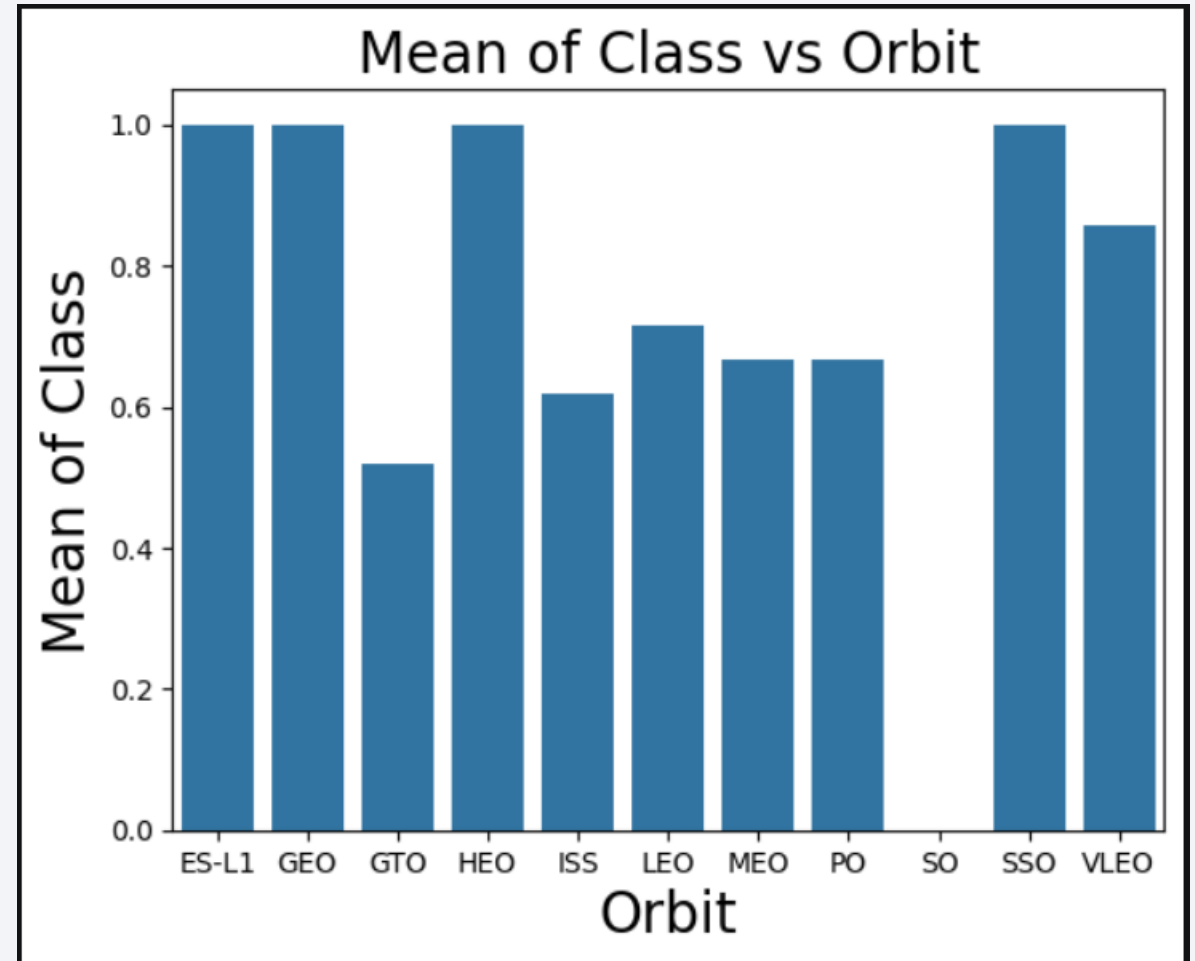
Payload vs. Launch Site

This graph shows the relationship between the payload mass and the launch site. This shows which launch sites typically launch rockets of a certain payload mass.



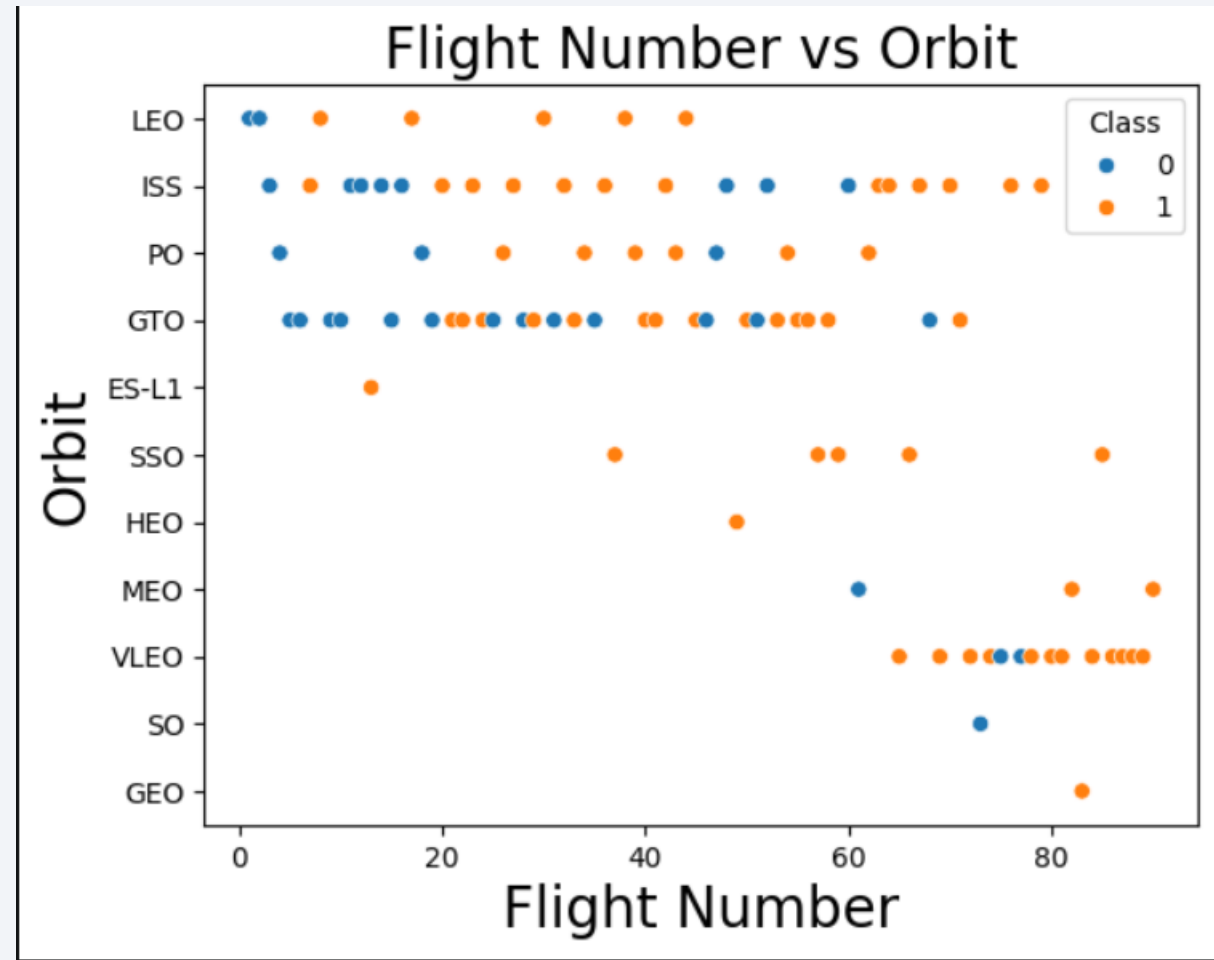
Success Rate vs. Orbit Type

This graph shows the relationship between the success rate and the orbit type. This is used to determine if launches going into different orbits are more or less likely to be a successful launch.



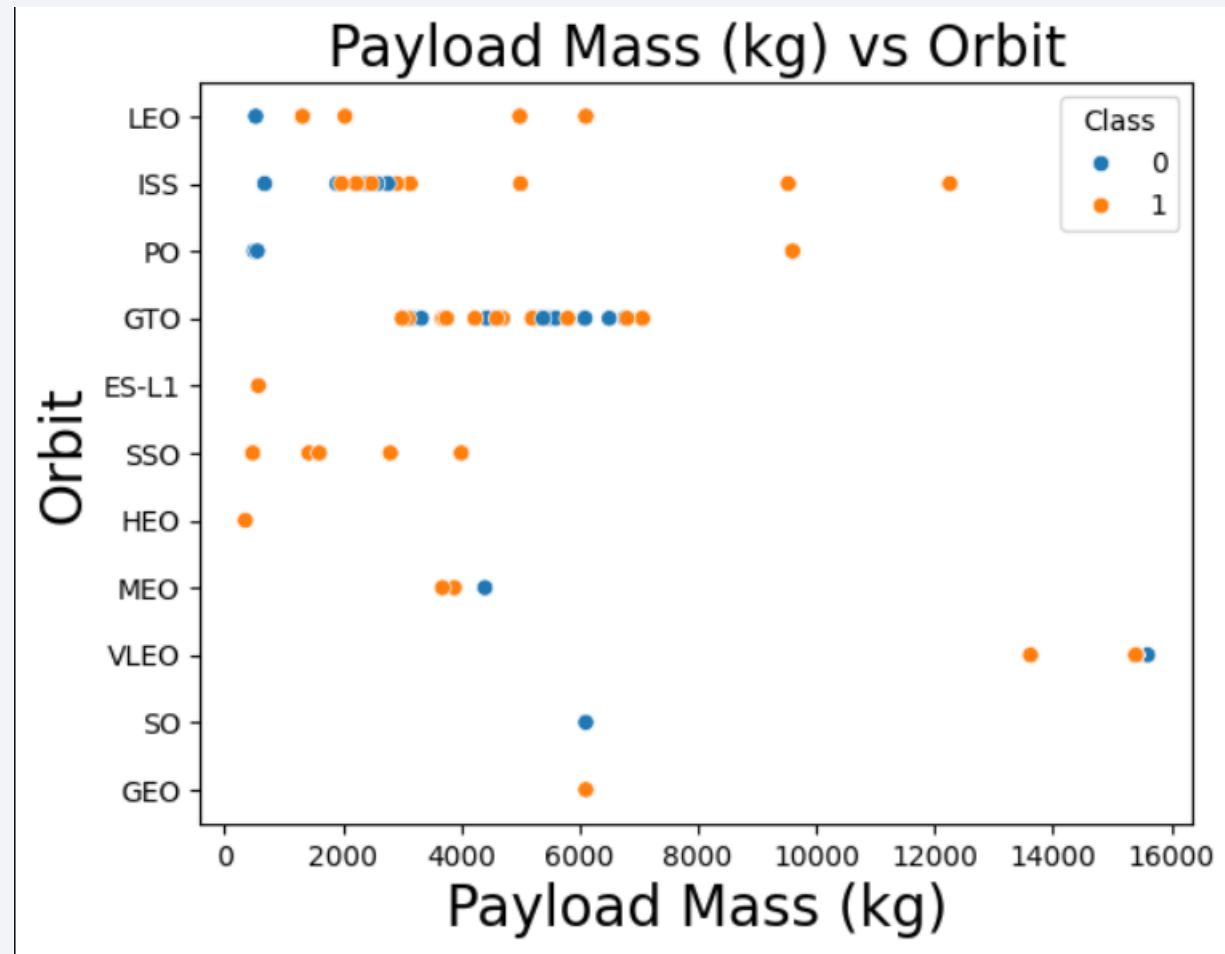
Flight Number vs. Orbit Type

This graph shows the relationship between flight number and orbit type. This is used to see the progression over time of which orbit types were used more frequently.



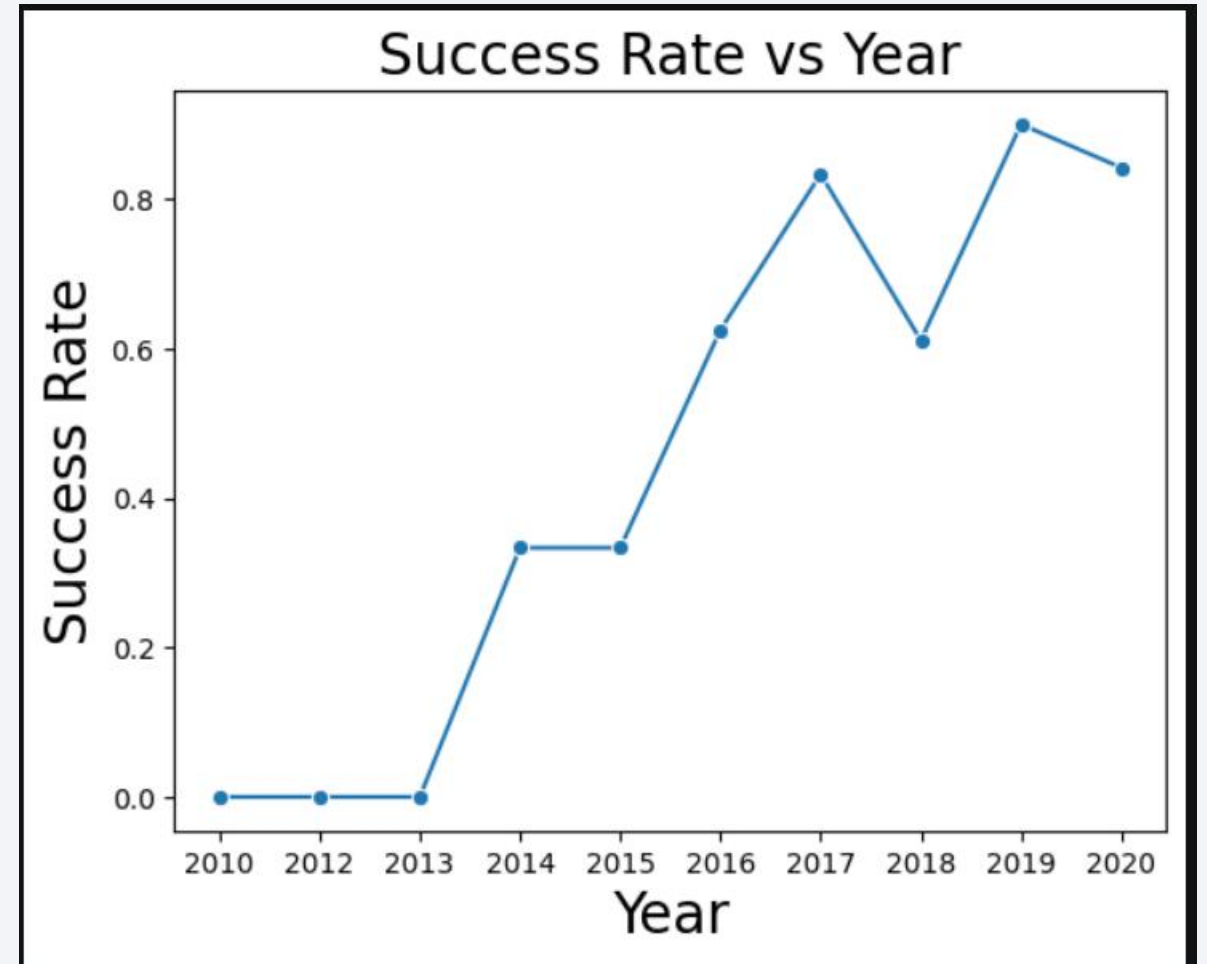
Payload vs. Orbit Type

This graph shows the relationship between payload mass and orbit type. This is used to determine if payloads within a certain range are commonly used for rockets going into a certain orbit.



Launch Success Yearly Trend

This graph shows the success rate of launches by year. This is used to see if the success rate overall increases over time likely due to better engineering and learning from previous mistakes.



All Launch Site Names

This query outputs each individual site name that has been used in a launch

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

This query outputs the first five instances of a launch coming from a launch site with a name beginning with 'CCA'. The CCA tag represents Cape Canaveral Florida so each launch site with this string attached to it is located within close proximity to eachother.

Total Payload Mass

Total_Payload_Mass

45596

This query shows the total pay load carried by each of the launches performed. It adds up the individual masses to get this total pay load mass.

Average Payload Mass by F9 v1.1

Average_Mass

2928.4

This query shows the average pay load mass used in an F9 v1.1 booster. This is to show how the pay load used in this particular booster type compares to other types and the pay loads they tend to carry.

First Successful Ground Landing Date

Min_Date
2015-12-22

This query shows the date of the first successful ground landing. This used to represent the start of the program so that we can see the progress that has been made since this date.

Successful Drone Ship Landing with Payload between 4000 and 6000

This query represents each of the boosters that have successfully landed with a payload weighing between 4000 and 6000 Kg. This is used to show the capabilities of these boosters for what payloads they can carry and successfully land with.

Booster_Version

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

Outcome_Category	Total_count
Failure	1
Success	100

This query lays out the total number of success and failure mission outcomes. This helps to understand how productive and successful the project has been.

Boosters Carried Maximum Payload

This query shows the booster versions that have carried the maximum payload mass used by the program. This is used to show which boosters tend to be used with the highest payload masses.

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

This query shows multiple pieces of information which as month, booster version, and launch site, for the landings with the outcome of failure on the drone ship in the year 2015. This is used to categorize these types of failures and show multiple different data points about them.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

This query shows each of the landing outcomes from the dates 2010-06-04 to 2017-03-20, ranking them by how frequently they occur. This helps to understand which outcomes tend to happen more often on these missions.

Landing_Outcome	Total_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

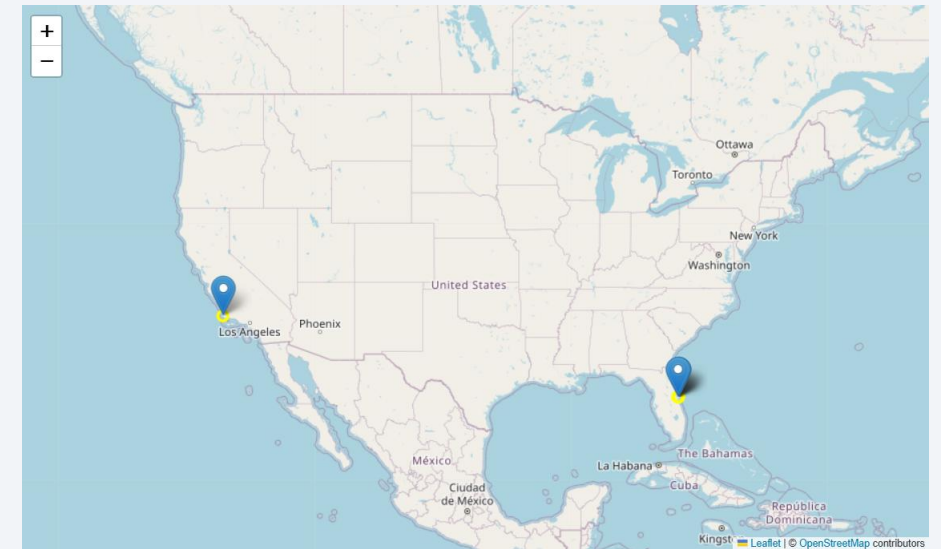
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

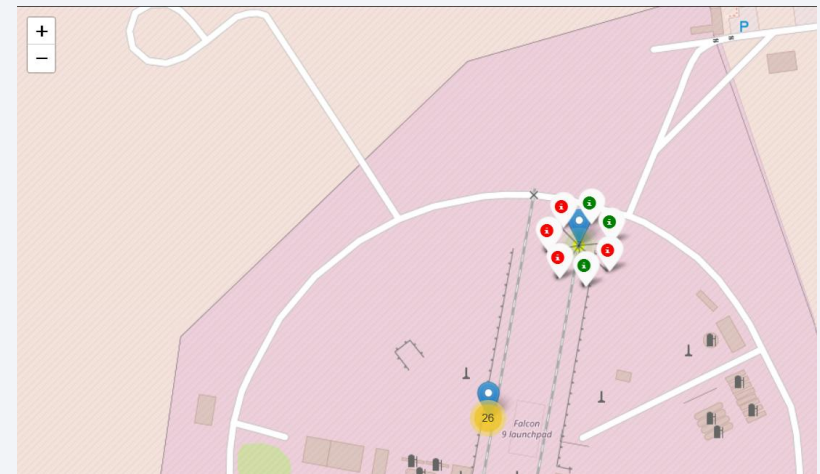
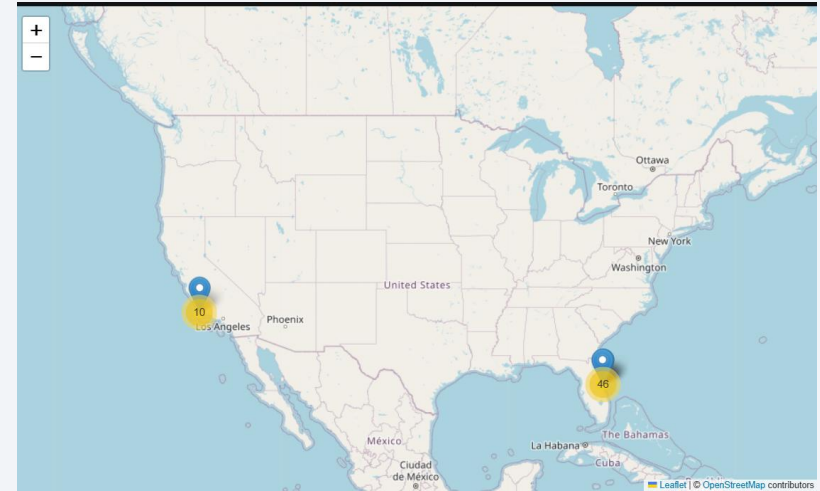
Map of Launch Sites

This map shows the locations of each of the launch sites across the country. From this map we can see that all launch sites are located on the coast either in Florida or California. The launch sites can be separated into two groups depending on which state they are from.



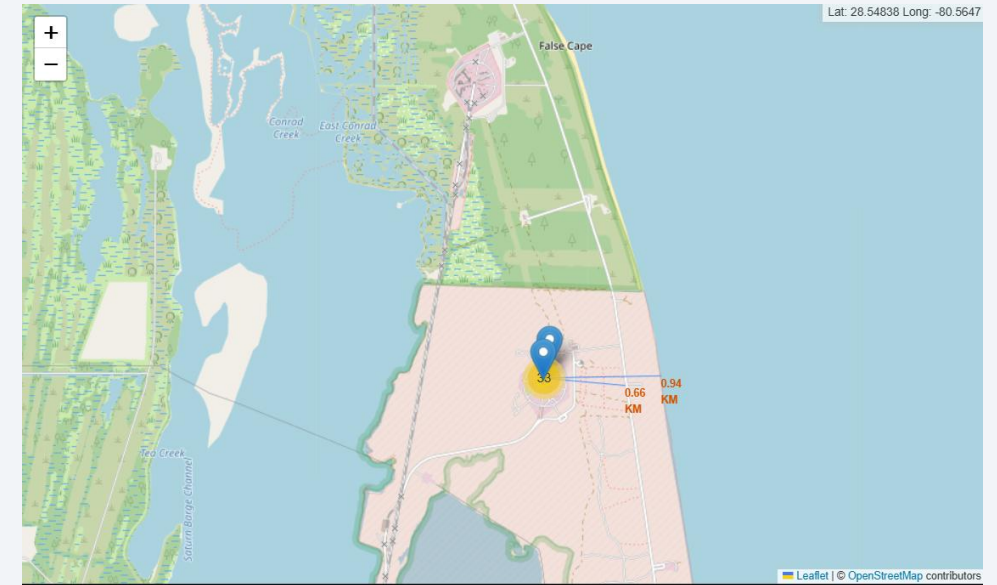
Map of Individual Launch Attempts

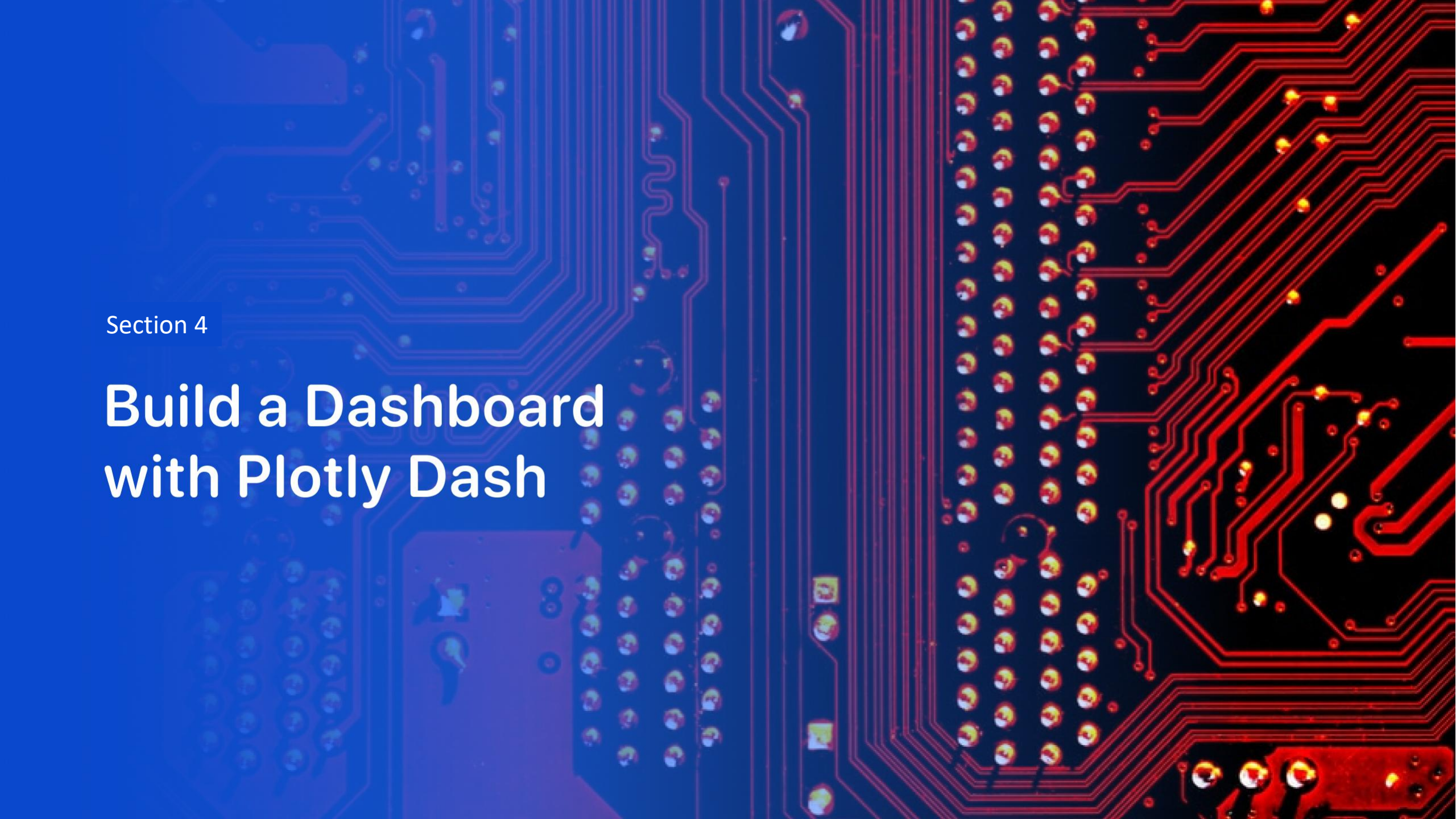
This map shows each launch attempt located at each launch site. When you zoom into a specific launch site you can see green or red markers indicating a successful or failed launch attempt, respectively. This helps to geographically pinpoint where most of the successes and failures take place and which launch sites have better success rates than others.



Launch Site Distance to Important Features

This map shows the distance between one launch site and important features surrounding the launch site that may have a significant impact on why the site was chosen for launches or the success rate of the launches. Through the distance labels we can see that the site is in very close proximity to both the shore and a highway. This is likely due to helping in the transportation of materials both to and from the launch site. The proximity to the coast also helps with the retrieval process as, after the booster lands on the water, there is a small distance left to travel to get to the launch site for it to be reused.



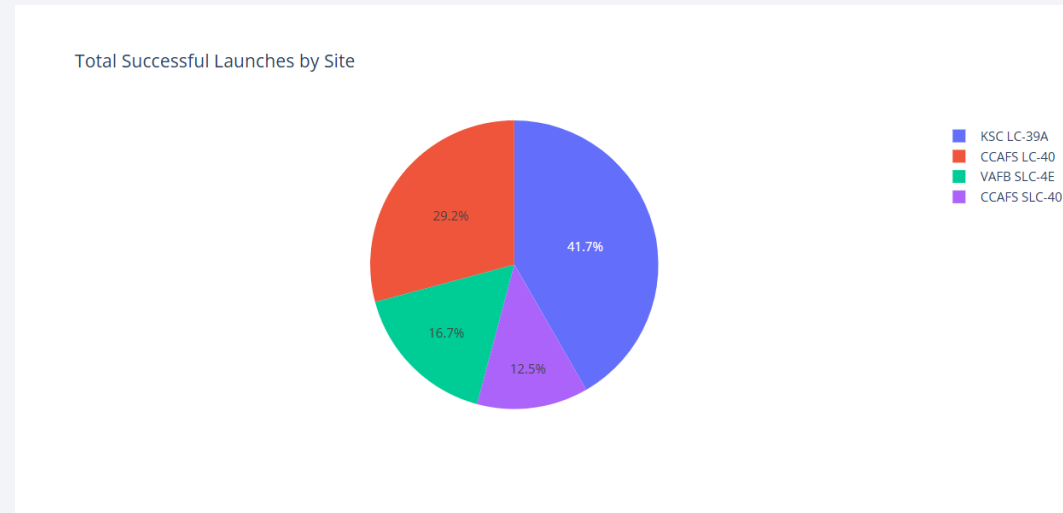


Section 4

Build a Dashboard with Plotly Dash

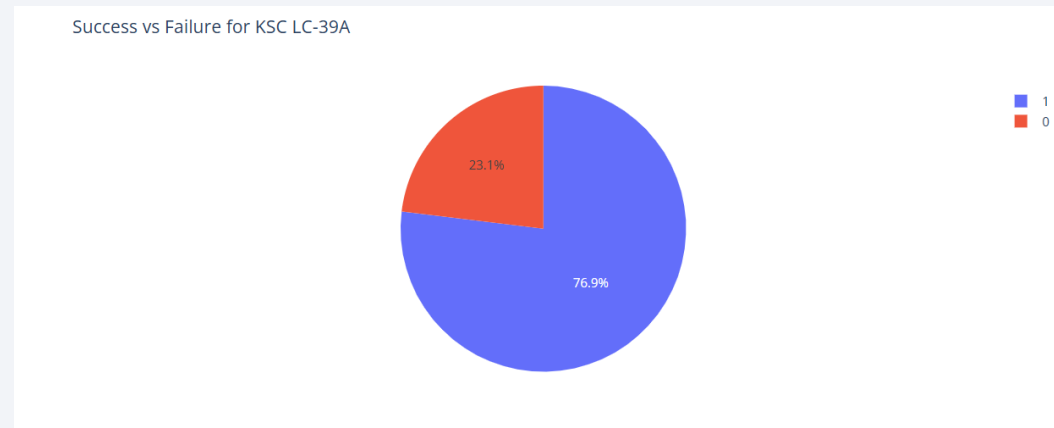
Total Successful Launches by Site

This pie chart shows the percentage, from the total number of successful launches, that have come from each site. From this chart we can see that KSC LC-39A has the highest percentage of the successful launches with CCAFS SLC-40 having the lowest percentage.



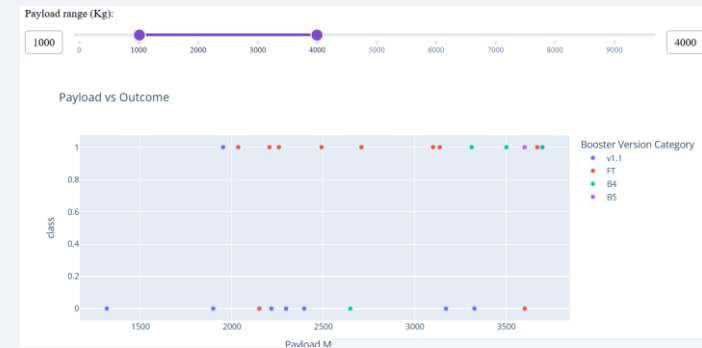
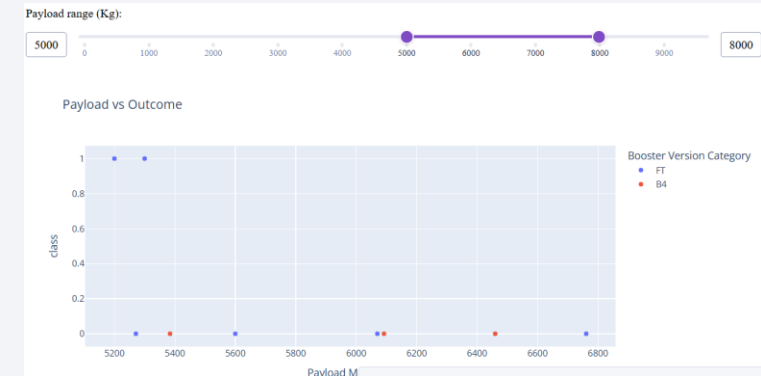
Success vs Failure for KSC LC-39A Launch Site

This pie chart shows the percentages of the launches coming from this launch site that were a success or a failure. This shows that 76.9% of the launches coming from this site were a success, and the remaining 23.1% were a failure.



Payload Range Scatterplots

This scatterplot compares the payload mass to the success of that mission. From this graph we can see that as the payload mass gets heavier, the success rate of the launch gets lower.

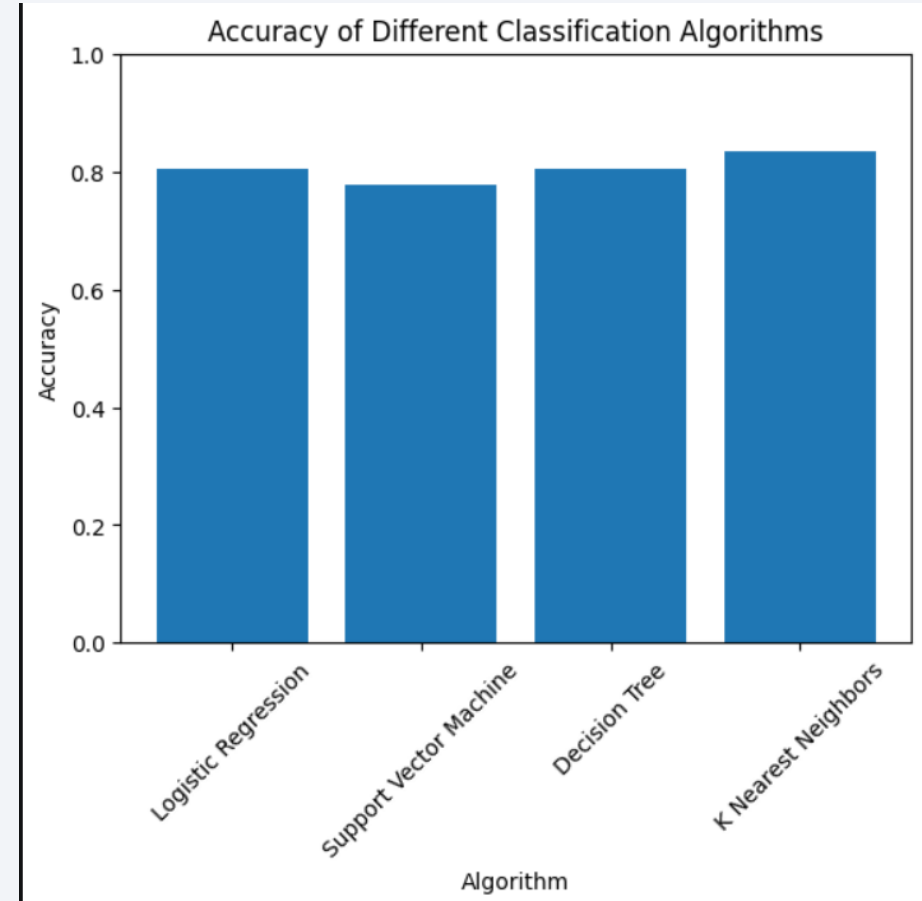


Section 5

Predictive Analysis (Classification)

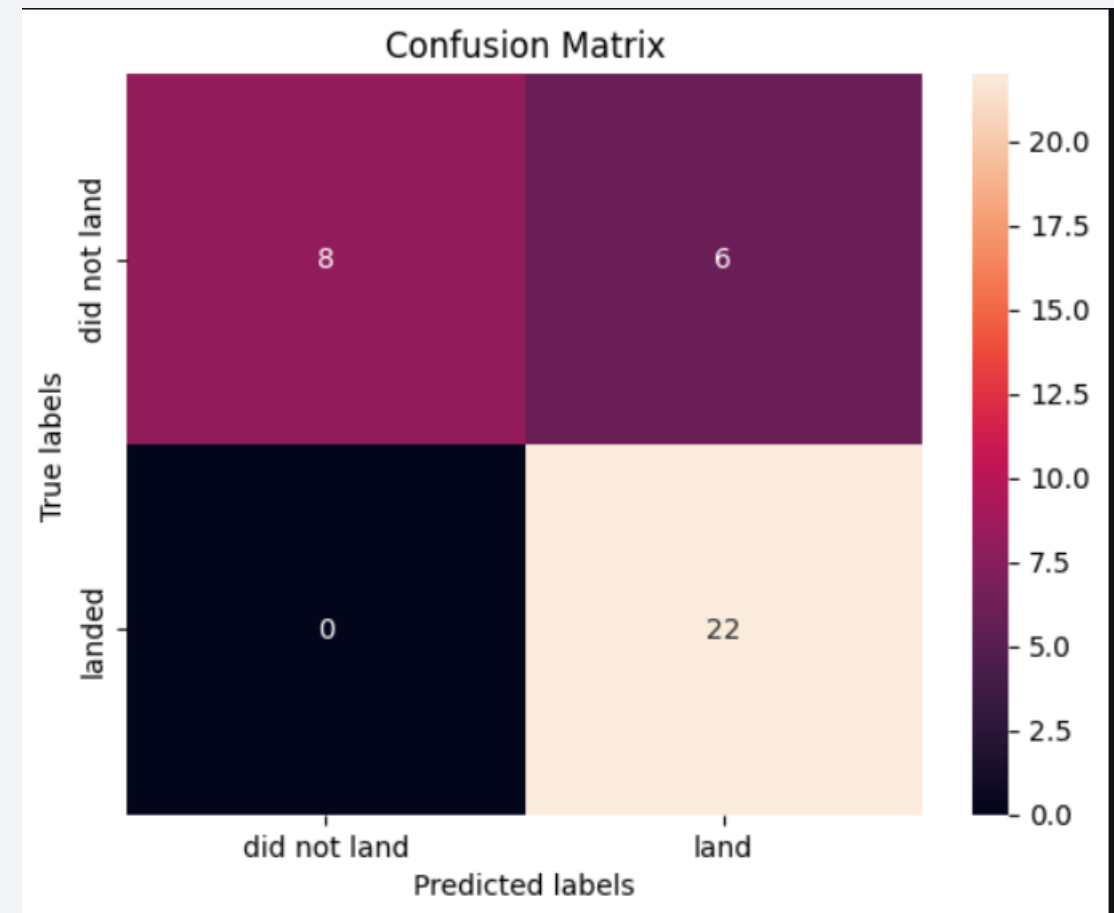
Classification Accuracy

This bar chart shows the accuracy for each of the used classification models. From this chart, we can see that the K Nearest Neighbors algorithm had the highest accuracy rate.



Confusion Matrix

This confusion matrix for the K Nearest Neighbors algorithm compares the predicted label with the true label for the dataset. The top left column represents a true negative, the top right represents a false positive, the bottom left represents a false negative, and the bottom right shows a true positive. The high number of true positive and true negative values shown in this matrix contributes to the idea that this model was very accurate in its classification.



Conclusions

Through this analysis we were able to determine that the success of the landing primarily depends on the payload mass, launch site, orbit type, and booster version. Using these data points, the K Nearest Neighbors classification algorithm was able to correctly identify the success of a launch at an 83% rate which was the most successful of the different models tested including logistic regression, support vector machine, and decision tree classification. The two most important predictors were found to be the orbit type and payload mass. Knowing these important factors and being able to predict the success of a launch is important because it will help add insight in preventing failed launch attempts. This will save money as SpaceX will not have to rebuild the rockets as frequently if parts can be spared in the launch process. Though the dataset was limited, potentially creating errors in the prediction model, the insight gained from this project can be used to effectively predict the landing outcome of Falcon 9 rockets and be used in the launch decision making process.

Thank you!

